

SIMULATION IDEAS

For each observation $i \in [n]$, denote by its tensor covariate $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \dots \times p_D}$, one-way covariates $\mathbf{z}_i \in \mathbb{R}^{p_0}$, and two responses $y_{ij} \in \mathbb{R}$ where $j = 1, 2$ represents two eyes. Let $t \in [0, 1]$ be the observation time, $\mathcal{A}(t) = (\alpha_{j_1, \dots, j_d}(t)) \in \mathbb{R}^{p_1 \times \dots \times p_D}$ be the unknown tensor coefficient function, and $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$ be the coefficients. The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \sum_{s=1}^S \left\langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

by a rank- R CP decomposition, we can reorganize it in a matrix form

$$\approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \right\rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right).$$

1 Paired Simulation for Case I

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p'_1 = p'_2 = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \quad \text{where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\alpha_{1rs}(t) = 5\sqrt{\frac{rs}{RS}}(t - 0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} - 0.75 \right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)), \quad (2)$$

$$\alpha_{4rs}(t) = \sqrt{\frac{rs}{RS}} \left[0.45 \cdot 5(t - 0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20 \sin(2\pi(t - 0.5)) \right]. \quad (3)$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. For now, we set $\sigma^2(t) \equiv 1$ as a toy model. The responses are generated by model eq(1).

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) of AOAS draft.

Simulation Results. We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Tables. All values in the following tables are reported in units of $\times 10^{-2}$; within each row, the better value between $\hat{\mathbf{A}}^\dagger(t)$ or $\hat{\mathbf{A}}^*(t)$ is highlighted.

- (i) When $\rho = 0$, there is almost no difference between iid estimation ($\hat{\mathbf{A}}^\dagger(t), \hat{\beta}^\dagger$) and paired estimation ($\hat{\mathbf{A}}^*(t), \hat{\beta}^*$).
- (ii) The larger the ρ is, the better the paired estimation $\hat{\mathbf{A}}^*(t)$ is than iid $\hat{\mathbf{A}}^\dagger(t)$.
- (iii) The larger the ρ is, the harder it is to estimate both β and the correlation ρ .
- (iv) All estimation errors generally decrease as n increases.

Table 1: Case 1 $\mathbf{A}(t)$ MIAE, β MAE, and covariance estimation errors.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.06 ($\pm$ 0.17)	5.08 ($\pm$ 0.18)	2.09 ($\pm$ 0.78)	2.12 ($\pm$ 0.77)	30.32 ($\pm$ 2.66)	2.81 ($\pm$ 1.91)
		0.3	5.05 ($\pm$ 0.20)	4.94 ($\pm$ 0.18)	2.44 ($\pm$ 1.01)	2.49 ($\pm$ 1.07)	30.23 ($\pm$ 3.48)	7.48 ($\pm$ 3.06)
		0.6	5.06 ($\pm$ 0.24)	4.53 ($\pm$ 0.17)	2.66 ($\pm$ 1.10)	2.69 ($\pm$ 1.13)	30.19 ($\pm$ 3.71)	15.09 ($\pm$ 2.55)
		0.9	5.06 ($\pm$ 0.27)	3.57 ($\pm$ 0.18)	2.88 ($\pm$ 1.13)	2.84 ($\pm$ 1.15)	30.14 ($\pm$ 3.99)	22.70 ($\pm$ 1.92)
	2000	0.0	3.29 ($\pm$ 0.10)	3.30 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.24 ($\pm$ 0.44)	13.31 ($\pm$ 1.88)	1.37 ($\pm$ 0.80)
		0.3	3.32 ($\pm$ 0.11)	3.22 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.36 ($\pm$ 0.61)	13.24 ($\pm$ 1.82)	4.12 ($\pm$ 1.64)
		0.6	3.33 ($\pm$ 0.11)	2.89 ($\pm$ 0.10)	1.53 ($\pm$ 0.65)	1.47 ($\pm$ 0.67)	13.15 ($\pm$ 2.04)	8.48 ($\pm$ 1.41)
		0.9	3.33 ($\pm$ 0.11)	2.16 ($\pm$ 0.07)	1.68 ($\pm$ 0.70)	1.56 ($\pm$ 0.72)	13.07 ($\pm$ 2.32)	12.92 ($\pm$ 1.03)
$\mathbf{A}_2(t)$	1000	0.0	5.08 ($\pm$ 0.21)	5.10 ($\pm$ 0.21)	2.09 ($\pm$ 0.79)	2.12 ($\pm$ 0.78)	30.33 ($\pm$ 2.72)	2.94 ($\pm$ 2.11)
		0.3	5.03 ($\pm$ 0.18)	4.92 ($\pm$ 0.16)	2.44 ($\pm$ 0.98)	2.51 ($\pm$ 1.03)	30.28 ($\pm$ 3.56)	6.87 ($\pm$ 2.92)
		0.6	5.03 ($\pm$ 0.20)	4.50 ($\pm$ 0.16)	2.66 ($\pm$ 1.08)	2.71 ($\pm$ 1.11)	30.25 ($\pm$ 3.78)	14.46 ($\pm$ 2.31)
		0.9	5.02 ($\pm$ 0.22)	3.52 ($\pm$ 0.15)	2.89 ($\pm$ 1.12)	2.84 ($\pm$ 1.14)	30.20 ($\pm$ 4.05)	22.09 ($\pm$ 1.48)
	2000	0.0	3.28 ($\pm$ 0.08)	3.29 ($\pm$ 0.08)	1.21 ($\pm$ 0.46)	1.21 ($\pm$ 0.46)	13.43 ($\pm$ 1.89)	1.56 ($\pm$ 0.96)
		0.3	3.29 ($\pm$ 0.12)	3.19 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.37 ($\pm$ 0.63)	13.34 ($\pm$ 1.78)	3.87 ($\pm$ 1.59)
		0.6	3.29 ($\pm$ 0.12)	2.86 ($\pm$ 0.10)	1.54 ($\pm$ 0.64)	1.48 ($\pm$ 0.70)	13.24 ($\pm$ 2.03)	8.22 ($\pm$ 1.31)
		0.9	3.29 ($\pm$ 0.11)	2.09 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.60 ($\pm$ 0.74)	13.12 ($\pm$ 2.34)	12.64 ($\pm$ 0.98)
$\mathbf{A}_3(t)$	1000	0.0	6.49 ($\pm$ 0.22)	6.60 ($\pm$ 0.23)	2.21 ($\pm$ 0.76)	2.27 ($\pm$ 0.76)	17.16 ($\pm$ 2.64)	2.85 ($\pm$ 2.21)
		0.3	6.52 ($\pm$ 0.18)	6.56 ($\pm$ 0.17)	2.46 ($\pm$ 1.11)	2.53 ($\pm$ 1.09)	17.08 ($\pm$ 3.21)	11.00 ($\pm$ 3.08)
		0.6	6.53 ($\pm$ 0.19)	6.29 ($\pm$ 0.16)	2.70 ($\pm$ 1.18)	2.73 ($\pm$ 1.16)	17.15 ($\pm$ 3.26)	22.32 ($\pm$ 2.77)
		0.9	6.54 ($\pm$ 0.22)	5.75 ($\pm$ 0.15)	2.95 ($\pm$ 1.21)	2.89 ($\pm$ 1.22)	17.24 ($\pm$ 3.35)	33.66 ($\pm$ 2.46)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.49 ($\pm$ 0.56)	1.48 ($\pm$ 0.58)	9.82 ($\pm$ 1.17)	1.33 ($\pm$ 0.98)
		0.3	5.05 ($\pm$ 0.12)	5.09 ($\pm$ 0.12)	1.41 ($\pm$ 0.59)	1.38 ($\pm$ 0.61)	10.29 ($\pm$ 1.30)	8.32 ($\pm$ 1.90)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.55 ($\pm$ 0.65)	1.49 ($\pm$ 0.66)	10.41 ($\pm$ 1.36)	16.81 ($\pm$ 1.83)
		0.9	5.06 ($\pm$ 0.13)	4.50 ($\pm$ 0.09)	1.67 ($\pm$ 0.70)	1.58 ($\pm$ 0.70)	10.57 ($\pm$ 1.48)	25.32 ($\pm$ 1.67)
$\mathbf{A}_4(t)$	1000	0.0	4.87 ($\pm$ 0.19)	4.88 ($\pm$ 0.20)	2.09 ($\pm$ 0.76)	2.13 ($\pm$ 0.75)	31.77 ($\pm$ 2.69)	2.85 ($\pm$ 1.97)
		0.3	4.85 ($\pm$ 0.18)	4.73 ($\pm$ 0.16)	2.43 ($\pm$ 1.01)	2.47 ($\pm$ 1.07)	31.68 ($\pm$ 3.47)	6.83 ($\pm$ 2.97)
		0.6	4.85 ($\pm$ 0.21)	4.28 ($\pm$ 0.15)	2.65 ($\pm$ 1.10)	2.67 ($\pm$ 1.14)	31.64 ($\pm$ 3.69)	13.98 ($\pm$ 2.37)
		0.9	4.86 ($\pm$ 0.24)	3.20 ($\pm$ 0.16)	2.88 ($\pm$ 1.13)	2.82 ($\pm$ 1.15)	31.59 ($\pm$ 3.97)	21.14 ($\pm$ 1.64)
	2000	0.0	3.02 ($\pm$ 0.09)	3.02 ($\pm$ 0.09)	1.26 ($\pm$ 0.45)	1.26 ($\pm$ 0.45)	15.12 ($\pm$ 1.87)	1.44 ($\pm$ 0.78)
		0.3	3.03 ($\pm$ 0.09)	2.91 ($\pm$ 0.09)	1.36 ($\pm$ 0.58)	1.33 ($\pm$ 0.60)	15.04 ($\pm$ 1.85)	3.57 ($\pm$ 1.45)
		0.6	3.04 ($\pm$ 0.09)	2.54 ($\pm$ 0.08)	1.51 ($\pm$ 0.64)	1.45 ($\pm$ 0.66)	14.95 ($\pm$ 2.06)	7.33 ($\pm$ 1.30)
		0.9	3.04 ($\pm$ 0.09)	1.63 ($\pm$ 0.06)	1.65 ($\pm$ 0.69)	1.54 ($\pm$ 0.71)	14.85 ($\pm$ 2.35)	11.21 ($\pm$ 0.88)

Table 2: Case 1 $\mathbf{A}(t)$ RMISE, β RMSE, and covariance estimation errors.

function	n	ρ	RMISE		RMSE		RMISE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	6.93 ($\pm$ 0.33)	6.97 ($\pm$ 0.34)	2.45 ($\pm$ 0.94)	2.50 ($\pm$ 0.98)	31.46 ($\pm$ 2.49)	2.81 ($\pm$ 1.91)
		0.3	6.95 ($\pm$ 0.38)	6.83 ($\pm$ 0.35)	2.78 ($\pm$ 1.07)	2.83 ($\pm$ 1.09)	31.34 ($\pm$ 3.35)	7.48 ($\pm$ 3.06)
		0.6	6.96 ($\pm$ 0.43)	6.30 ($\pm$ 0.36)	3.05 ($\pm$ 1.16)	3.04 ($\pm$ 1.17)	31.33 ($\pm$ 3.57)	15.09 ($\pm$ 2.55)
		0.9	6.96 ($\pm$ 0.50)	5.09 ($\pm$ 0.40)	3.32 ($\pm$ 1.23)	3.19 ($\pm$ 1.22)	31.34 ($\pm$ 3.82)	22.70 ($\pm$ 1.92)
	2000	0.0	4.28 ($\pm$ 0.14)	4.28 ($\pm$ 0.14)	1.45 ($\pm$ 0.49)	1.44 ($\pm$ 0.49)	14.39 ($\pm$ 1.87)	1.37 ($\pm$ 0.80)
		0.3	4.33 ($\pm$ 0.14)	4.21 ($\pm$ 0.15)	1.62 ($\pm$ 0.64)	1.58 ($\pm$ 0.67)	14.54 ($\pm$ 1.74)	4.12 ($\pm$ 1.64)
		0.6	4.34 ($\pm$ 0.14)	3.77 ($\pm$ 0.13)	1.79 ($\pm$ 0.70)	1.71 ($\pm$ 0.74)	14.57 ($\pm$ 1.98)	8.48 ($\pm$ 1.41)
		0.9	4.34 ($\pm$ 0.14)	2.74 ($\pm$ 0.09)	1.95 ($\pm$ 0.75)	1.82 ($\pm$ 0.77)	14.66 ($\pm$ 2.26)	12.92 ($\pm$ 1.03)
$\mathbf{A}_2(t)$	1000	0.0	6.94 ($\pm$ 0.36)	6.98 ($\pm$ 0.37)	2.47 ($\pm$ 0.95)	2.51 ($\pm$ 0.98)	31.57 ($\pm$ 2.57)	2.94 ($\pm$ 2.11)
		0.3	6.92 ($\pm$ 0.36)	6.82 ($\pm$ 0.34)	2.81 ($\pm$ 1.02)	2.85 ($\pm$ 1.05)	31.47 ($\pm$ 3.40)	6.87 ($\pm$ 2.92)
		0.6	6.93 ($\pm$ 0.41)	6.28 ($\pm$ 0.35)	3.08 ($\pm$ 1.11)	3.05 ($\pm$ 1.14)	31.46 ($\pm$ 3.60)	14.46 ($\pm$ 2.31)
		0.9	6.92 ($\pm$ 0.47)	5.04 ($\pm$ 0.37)	3.34 ($\pm$ 1.18)	3.20 ($\pm$ 1.20)	31.47 ($\pm$ 3.84)	22.09 ($\pm$ 1.48)
	2000	0.0	4.27 ($\pm$ 0.10)	4.28 ($\pm$ 0.11)	1.43 ($\pm$ 0.53)	1.43 ($\pm$ 0.53)	14.68 ($\pm$ 1.89)	1.56 ($\pm$ 0.96)
		0.3	4.31 ($\pm$ 0.15)	4.19 ($\pm$ 0.15)	1.63 ($\pm$ 0.65)	1.60 ($\pm$ 0.70)	14.77 ($\pm$ 1.74)	3.87 ($\pm$ 1.59)
		0.6	4.31 ($\pm$ 0.15)	3.75 ($\pm$ 0.13)	1.81 ($\pm$ 0.70)	1.73 ($\pm$ 0.77)	14.77 ($\pm$ 2.01)	8.22 ($\pm$ 1.31)
		0.9	4.31 ($\pm$ 0.15)	2.71 ($\pm$ 0.10)	1.98 ($\pm$ 0.74)	1.85 ($\pm$ 0.81)	14.82 ($\pm$ 2.32)	12.64 ($\pm$ 0.98)
$\mathbf{A}_3(t)$	1000	0.0	8.41 ($\pm$ 0.34)	8.55 ($\pm$ 0.35)	2.64 ($\pm$ 0.91)	2.71 ($\pm$ 0.93)	20.98 ($\pm$ 2.18)	2.85 ($\pm$ 2.21)
		0.3	8.46 ($\pm$ 0.28)	8.50 ($\pm$ 0.27)	2.89 ($\pm$ 1.17)	2.97 ($\pm$ 1.10)	20.88 ($\pm$ 3.01)	11.00 ($\pm$ 3.08)
		0.6	8.47 ($\pm$ 0.32)	8.15 ($\pm$ 0.27)	3.14 ($\pm$ 1.26)	3.17 ($\pm$ 1.20)	20.87 ($\pm$ 3.13)	22.32 ($\pm$ 2.77)
		0.9	8.48 ($\pm$ 0.38)	7.45 ($\pm$ 0.28)	3.40 ($\pm$ 1.32)	3.33 ($\pm$ 1.28)	20.90 ($\pm$ 3.26)	33.66 ($\pm$ 2.46)
	2000	0.0	6.25 ($\pm$ 0.12)	6.38 ($\pm$ 0.13)	1.75 ($\pm$ 0.59)	1.74 ($\pm$ 0.59)	11.69 ($\pm$ 1.32)	1.33 ($\pm$ 0.98)
		0.3	6.27 ($\pm$ 0.15)	6.32 ($\pm$ 0.15)	1.63 ($\pm$ 0.68)	1.60 ($\pm$ 0.70)	12.23 ($\pm$ 1.49)	8.32 ($\pm$ 1.90)
		0.6	6.28 ($\pm$ 0.15)	6.07 ($\pm$ 0.14)	1.78 ($\pm$ 0.74)	1.72 ($\pm$ 0.76)	12.39 ($\pm$ 1.58)	16.81 ($\pm$ 1.83)
		0.9	6.28 ($\pm$ 0.16)	5.59 ($\pm$ 0.11)	1.92 ($\pm$ 0.80)	1.83 ($\pm$ 0.79)	12.61 ($\pm$ 1.73)	25.32 ($\pm$ 1.67)
$\mathbf{A}_4(t)$	1000	0.0	6.75 ($\pm$ 0.36)	6.78 ($\pm$ 0.36)	2.45 ($\pm$ 0.93)	2.50 ($\pm$ 0.95)	32.74 ($\pm$ 2.54)	2.85 ($\pm$ 1.97)
		0.3	6.76 ($\pm$ 0.36)	6.64 ($\pm$ 0.33)	2.78 ($\pm$ 1.06)	2.82 ($\pm$ 1.07)	32.62 ($\pm$ 3.35)	6.83 ($\pm$ 2.97)
		0.6	6.77 ($\pm$ 0.41)	6.08 ($\pm$ 0.35)	3.05 ($\pm$ 1.15)	3.03 ($\pm$ 1.16)	32.61 ($\pm$ 3.56)	13.98 ($\pm$ 2.37)
		0.9	6.77 ($\pm$ 0.48)	4.76 ($\pm$ 0.39)	3.32 ($\pm$ 1.22)	3.17 ($\pm$ 1.22)	32.61 ($\pm$ 3.81)	21.14 ($\pm$ 1.64)
	2000	0.0	4.00 ($\pm$ 0.12)	4.00 ($\pm$ 0.13)	1.46 ($\pm$ 0.50)	1.46 ($\pm$ 0.50)	15.89 ($\pm$ 1.86)	1.44 ($\pm$ 0.78)
		0.3	4.05 ($\pm$ 0.13)	3.90 ($\pm$ 0.14)	1.58 ($\pm$ 0.63)	1.55 ($\pm$ 0.66)	15.99 ($\pm$ 1.75)	3.57 ($\pm$ 1.45)
		0.6	4.05 ($\pm$ 0.13)	3.42 ($\pm$ 0.12)	1.75 ($\pm$ 0.69)	1.68 ($\pm$ 0.74)	16.01 ($\pm$ 2.00)	7.33 ($\pm$ 1.30)
		0.9	4.05 ($\pm$ 0.13)	2.21 ($\pm$ 0.09)	1.92 ($\pm$ 0.74)	1.79 ($\pm$ 0.77)	16.06 ($\pm$ 2.30)	11.21 ($\pm$ 0.88)

2 Paired Simulation for Case II

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 48$, and $S = 27$. We split the tensor by $S^{(1)} = S^{(2)} = S^{(3)} = 3$ and $p'_1 = p'_2 = p'_3 = 16$. Set the rank $R = 6$, and covariate tensor:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has 4 choices in Eq(2) and (3). Everything else is same as Case I.

Simulation Results. We vary $n = 1000/2000$, coefficients $\mathcal{A}_1(t), \dots, \mathcal{A}_4(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\beta}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\beta}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Tables. All values in the following tables are reported in units of $\times 10^{-2}$; within each row, the better value between $\hat{\mathbf{A}}^\dagger(t)$ or $\hat{\mathbf{A}}^*(t)$ is highlighted.

- (i) When $\rho = 0$, there is almost no difference between iid estimation $(\hat{\mathbf{A}}^\dagger(t), \hat{\beta}^\dagger)$ and paired estimation $(\hat{\mathbf{A}}^*(t), \hat{\beta}^*)$.
- (ii) The larger the ρ is, the better the paired estimation $\hat{\mathbf{A}}^*(t)$ is than iid $\hat{\mathbf{A}}^\dagger(t)$.
- (iii) The larger the ρ is, the harder it is to estimate both β and the correlation ρ .
- (iv) All estimation errors generally decrease as n increases.

Table 3: Case 2 $\mathbf{A}(t)$ MIAE, β MAE, and covariance estimation errors.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.51 ($\pm$ 0.10)	3.53 ($\pm$ 0.10)	1.55 ($\pm$ 0.65)	1.53 ($\pm$ 0.66)	18.86 ($\pm$ 2.34)	1.96 ($\pm$ 1.45)
		0.3	3.51 ($\pm$ 0.10)	3.44 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.69 ($\pm$ 0.71)	18.64 ($\pm$ 2.18)	6.54 ($\pm$ 2.61)
		0.6	3.51 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.86 ($\pm$ 0.77)	1.83 ($\pm$ 0.75)	18.61 ($\pm$ 2.62)	13.45 ($\pm$ 2.06)
		0.9	3.52 ($\pm$ 0.11)	2.58 ($\pm$ 0.08)	2.02 ($\pm$ 0.83)	1.96 ($\pm$ 0.77)	18.66 ($\pm$ 2.97)	20.39 ($\pm$ 1.34)
	5000	0.0	2.42 ($\pm$ 0.06)	2.43 ($\pm$ 0.06)	0.90 ($\pm$ 0.36)	0.91 ($\pm$ 0.37)	4.58 ($\pm$ 0.92)	1.15 ($\pm$ 0.78)
		0.3	2.39 ($\pm$ 0.06)	2.35 ($\pm$ 0.06)	0.95 ($\pm$ 0.30)	0.94 ($\pm$ 0.30)	4.61 ($\pm$ 0.98)	3.99 ($\pm$ 1.29)
		0.6	2.39 ($\pm$ 0.06)	2.20 ($\pm$ 0.06)	1.04 ($\pm$ 0.34)	1.03 ($\pm$ 0.32)	4.78 ($\pm$ 1.16)	7.73 ($\pm$ 1.04)
		0.9	2.39 ($\pm$ 0.06)	1.89 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.11 ($\pm$ 0.34)	5.03 ($\pm$ 1.35)	11.42 ($\pm$ 0.64)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.52 ($\pm$ 0.72)	1.50 ($\pm$ 0.73)	19.09 ($\pm$ 2.34)	1.68 ($\pm$ 1.37)
		0.3	3.45 ($\pm$ 0.09)	3.39 ($\pm$ 0.10)	1.75 ($\pm$ 0.68)	1.73 ($\pm$ 0.67)	18.90 ($\pm$ 2.18)	6.28 ($\pm$ 2.39)
		0.6	3.45 ($\pm$ 0.09)	3.11 ($\pm$ 0.09)	1.91 ($\pm$ 0.74)	1.87 ($\pm$ 0.74)	18.89 ($\pm$ 2.55)	13.04 ($\pm$ 2.10)
		0.9	3.45 ($\pm$ 0.10)	2.47 ($\pm$ 0.08)	2.06 ($\pm$ 0.80)	1.99 ($\pm$ 0.78)	18.92 ($\pm$ 2.91)	19.93 ($\pm$ 1.51)
	5000	0.0	2.35 ($\pm$ 0.05)	2.36 ($\pm$ 0.05)	0.91 ($\pm$ 0.38)	0.91 ($\pm$ 0.39)	6.60 ($\pm$ 0.93)	1.21 ($\pm$ 0.73)
		0.3	2.33 ($\pm$ 0.05)	2.29 ($\pm$ 0.05)	0.92 ($\pm$ 0.30)	0.91 ($\pm$ 0.29)	6.61 ($\pm$ 0.96)	3.91 ($\pm$ 1.23)
		0.6	2.32 ($\pm$ 0.05)	2.12 ($\pm$ 0.04)	1.02 ($\pm$ 0.32)	1.00 ($\pm$ 0.31)	6.73 ($\pm$ 1.09)	7.63 ($\pm$ 0.99)
		0.9	2.32 ($\pm$ 0.05)	1.74 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.33)	6.88 ($\pm$ 1.27)	11.32 ($\pm$ 0.66)
$\mathbf{A}_3(t)$	2000	0.0	5.71 ($\pm$ 0.10)	5.91 ($\pm$ 0.10)	1.69 ($\pm$ 0.66)	1.63 ($\pm$ 0.65)	19.93 ($\pm$ 2.11)	1.87 ($\pm$ 1.71)
		0.3	5.71 ($\pm$ 0.11)	5.87 ($\pm$ 0.11)	1.87 ($\pm$ 0.86)	1.84 ($\pm$ 0.89)	20.13 ($\pm$ 2.11)	13.90 ($\pm$ 2.51)
		0.6	5.72 ($\pm$ 0.11)	5.73 ($\pm$ 0.11)	2.02 ($\pm$ 0.93)	1.97 ($\pm$ 0.96)	20.22 ($\pm$ 2.45)	27.57 ($\pm$ 2.09)
		0.9	5.73 ($\pm$ 0.11)	5.47 ($\pm$ 0.10)	2.16 ($\pm$ 0.99)	2.08 ($\pm$ 1.02)	20.31 ($\pm$ 2.82)	41.25 ($\pm$ 1.62)
	5000	0.0	4.90 ($\pm$ 0.06)	5.11 ($\pm$ 0.06)	1.03 ($\pm$ 0.36)	1.03 ($\pm$ 0.37)	36.89 ($\pm$ 1.55)	1.02 ($\pm$ 0.75)
		0.3	4.90 ($\pm$ 0.07)	5.08 ($\pm$ 0.07)	1.04 ($\pm$ 0.44)	1.04 ($\pm$ 0.43)	36.94 ($\pm$ 1.76)	11.98 ($\pm$ 1.32)
		0.6	4.90 ($\pm$ 0.07)	5.00 ($\pm$ 0.06)	1.13 ($\pm$ 0.46)	1.13 ($\pm$ 0.46)	36.85 ($\pm$ 1.89)	23.54 ($\pm$ 1.24)
		0.9	4.91 ($\pm$ 0.06)	4.84 ($\pm$ 0.05)	1.21 ($\pm$ 0.49)	1.21 ($\pm$ 0.48)	36.75 ($\pm$ 2.08)	35.04 ($\pm$ 1.14)
$\mathbf{A}_4(t)$	2000	0.0	3.13 ($\pm$ 0.09)	3.14 ($\pm$ 0.09)	1.51 ($\pm$ 0.67)	1.49 ($\pm$ 0.67)	22.56 ($\pm$ 2.39)	1.83 ($\pm$ 1.46)
		0.3	3.12 ($\pm$ 0.08)	3.03 ($\pm$ 0.08)	1.68 ($\pm$ 0.70)	1.67 ($\pm$ 0.71)	22.38 ($\pm$ 2.22)	5.44 ($\pm$ 2.37)
		0.6	3.13 ($\pm$ 0.09)	2.71 ($\pm$ 0.07)	1.85 ($\pm$ 0.77)	1.81 ($\pm$ 0.77)	22.33 ($\pm$ 2.70)	11.14 ($\pm$ 1.88)
		0.9	3.13 ($\pm$ 0.10)	1.91 ($\pm$ 0.06)	2.00 ($\pm$ 0.82)	1.94 ($\pm$ 0.80)	22.29 ($\pm$ 3.24)	16.95 ($\pm$ 1.02)
	5000	0.0	1.87 ($\pm$ 0.04)	1.87 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.91 ($\pm$ 0.34)	8.40 ($\pm$ 1.33)	1.13 ($\pm$ 0.74)
		0.3	1.84 ($\pm$ 0.05)	1.78 ($\pm$ 0.05)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	8.34 ($\pm$ 1.30)	2.81 ($\pm$ 1.20)
		0.6	1.84 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.32)	8.43 ($\pm$ 1.43)	5.30 ($\pm$ 0.92)
		0.9	1.84 ($\pm$ 0.06)	1.06 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.10 ($\pm$ 0.34)	8.52 ($\pm$ 1.65)	7.72 ($\pm$ 0.48)

Table 4: Case 2 $\mathbf{A}(t)$ RMISE, β RMSE, and covariance estimation errors.

function	n	ρ	RMISE		RMSE		RMISE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	4.59 ($\pm$ 0.13)	4.61 ($\pm$ 0.13)	1.85 ($\pm$ 0.77)	1.84 ($\pm$ 0.77)	20.35 ($\pm$ 2.06)	1.96 ($\pm$ 1.45)
		0.3	4.58 ($\pm$ 0.15)	4.50 ($\pm$ 0.14)	2.04 ($\pm$ 0.86)	2.01 ($\pm$ 0.83)	20.21 ($\pm$ 1.81)	6.54 ($\pm$ 2.61)
		0.6	4.59 ($\pm$ 0.15)	4.13 ($\pm$ 0.13)	2.25 ($\pm$ 0.94)	2.18 ($\pm$ 0.88)	20.26 ($\pm$ 2.15)	13.45 ($\pm$ 2.06)
		0.9	4.60 ($\pm$ 0.16)	3.30 ($\pm$ 0.11)	2.43 ($\pm$ 1.01)	2.34 ($\pm$ 0.89)	20.34 ($\pm$ 2.55)	20.39 ($\pm$ 1.34)
	5000	0.0	3.01 ($\pm$ 0.07)	3.02 ($\pm$ 0.07)	1.08 ($\pm$ 0.41)	1.08 ($\pm$ 0.42)	6.00 ($\pm$ 1.02)	1.15 ($\pm$ 0.78)
		0.3	2.97 ($\pm$ 0.07)	2.92 ($\pm$ 0.07)	1.11 ($\pm$ 0.35)	1.11 ($\pm$ 0.34)	6.09 ($\pm$ 1.09)	3.99 ($\pm$ 1.29)
		0.6	2.97 ($\pm$ 0.07)	2.71 ($\pm$ 0.06)	1.23 ($\pm$ 0.39)	1.21 ($\pm$ 0.37)	6.26 ($\pm$ 1.27)	7.73 ($\pm$ 1.04)
		0.9	2.97 ($\pm$ 0.07)	2.26 ($\pm$ 0.04)	1.33 ($\pm$ 0.43)	1.31 ($\pm$ 0.40)	6.48 ($\pm$ 1.52)	11.42 ($\pm$ 0.64)
$\mathbf{A}_2(t)$	2000	0.0	4.55 ($\pm$ 0.13)	4.58 ($\pm$ 0.13)	1.80 ($\pm$ 0.85)	1.79 ($\pm$ 0.84)	21.05 ($\pm$ 1.99)	1.68 ($\pm$ 1.37)
		0.3	4.54 ($\pm$ 0.14)	4.46 ($\pm$ 0.15)	2.07 ($\pm$ 0.80)	2.03 ($\pm$ 0.78)	20.86 ($\pm$ 1.79)	6.28 ($\pm$ 2.39)
		0.6	4.54 ($\pm$ 0.14)	4.10 ($\pm$ 0.14)	2.27 ($\pm$ 0.88)	2.20 ($\pm$ 0.84)	20.92 ($\pm$ 2.11)	13.04 ($\pm$ 2.10)
		0.9	4.54 ($\pm$ 0.14)	3.26 ($\pm$ 0.11)	2.45 ($\pm$ 0.96)	2.35 ($\pm$ 0.87)	21.03 ($\pm$ 2.49)	19.93 ($\pm$ 1.51)
	5000	0.0	2.98 ($\pm$ 0.06)	3.00 ($\pm$ 0.06)	1.10 ($\pm$ 0.43)	1.10 ($\pm$ 0.44)	7.69 ($\pm$ 1.00)	1.21 ($\pm$ 0.73)
		0.3	2.96 ($\pm$ 0.06)	2.92 ($\pm$ 0.06)	1.08 ($\pm$ 0.32)	1.07 ($\pm$ 0.32)	7.74 ($\pm$ 1.03)	3.91 ($\pm$ 1.23)
		0.6	2.96 ($\pm$ 0.06)	2.70 ($\pm$ 0.05)	1.19 ($\pm$ 0.36)	1.17 ($\pm$ 0.35)	7.90 ($\pm$ 1.17)	7.63 ($\pm$ 0.99)
		0.9	2.95 ($\pm$ 0.06)	2.25 ($\pm$ 0.04)	1.30 ($\pm$ 0.40)	1.27 ($\pm$ 0.37)	8.10 ($\pm$ 1.39)	11.32 ($\pm$ 0.66)
$\mathbf{A}_3(t)$	2000	0.0	7.13 ($\pm$ 0.11)	7.40 ($\pm$ 0.11)	1.96 ($\pm$ 0.80)	1.93 ($\pm$ 0.80)	23.61 ($\pm$ 2.31)	1.87 ($\pm$ 1.71)
		0.3	7.14 ($\pm$ 0.12)	7.34 ($\pm$ 0.12)	2.24 ($\pm$ 0.98)	2.19 ($\pm$ 1.01)	23.74 ($\pm$ 2.26)	13.90 ($\pm$ 2.51)
		0.6	7.14 ($\pm$ 0.12)	7.18 ($\pm$ 0.12)	2.42 ($\pm$ 1.07)	2.35 ($\pm$ 1.09)	23.81 ($\pm$ 2.65)	27.57 ($\pm$ 2.09)
		0.9	7.15 ($\pm$ 0.13)	6.87 ($\pm$ 0.11)	2.58 ($\pm$ 1.16)	2.49 ($\pm$ 1.16)	23.88 ($\pm$ 3.09)	41.25 ($\pm$ 1.62)
	5000	0.0	6.06 ($\pm$ 0.06)	6.35 ($\pm$ 0.06)	1.24 ($\pm$ 0.40)	1.23 ($\pm$ 0.41)	41.14 ($\pm$ 1.59)	1.02 ($\pm$ 0.75)
		0.3	6.06 ($\pm$ 0.08)	6.32 ($\pm$ 0.08)	1.23 ($\pm$ 0.49)	1.21 ($\pm$ 0.49)	41.17 ($\pm$ 1.80)	11.98 ($\pm$ 1.32)
		0.6	6.07 ($\pm$ 0.08)	6.24 ($\pm$ 0.07)	1.33 ($\pm$ 0.53)	1.32 ($\pm$ 0.52)	41.07 ($\pm$ 1.96)	23.54 ($\pm$ 1.24)
		0.9	6.07 ($\pm$ 0.08)	6.10 ($\pm$ 0.06)	1.43 ($\pm$ 0.57)	1.42 ($\pm$ 0.56)	40.97 ($\pm$ 2.17)	35.04 ($\pm$ 1.14)
$\mathbf{A}_4(t)$	2000	0.0	4.22 ($\pm$ 0.12)	4.23 ($\pm$ 0.12)	1.79 ($\pm$ 0.78)	1.78 ($\pm$ 0.78)	23.47 ($\pm$ 2.23)	1.83 ($\pm$ 1.46)
		0.3	4.21 ($\pm$ 0.14)	4.10 ($\pm$ 0.13)	2.03 ($\pm$ 0.85)	2.00 ($\pm$ 0.83)	23.31 ($\pm$ 1.97)	5.44 ($\pm$ 2.37)
		0.6	4.21 ($\pm$ 0.14)	3.68 ($\pm$ 0.12)	2.24 ($\pm$ 0.93)	2.17 ($\pm$ 0.89)	23.34 ($\pm$ 2.34)	11.14 ($\pm$ 1.88)
		0.9	4.22 ($\pm$ 0.15)	2.66 ($\pm$ 0.11)	2.42 ($\pm$ 1.01)	2.32 ($\pm$ 0.91)	23.40 ($\pm$ 2.76)	16.95 ($\pm$ 1.02)
	5000	0.0	2.44 ($\pm$ 0.06)	2.44 ($\pm$ 0.06)	1.08 ($\pm$ 0.38)	1.08 ($\pm$ 0.38)	9.03 ($\pm$ 1.29)	1.13 ($\pm$ 0.74)
		0.3	2.41 ($\pm$ 0.07)	2.33 ($\pm$ 0.07)	1.08 ($\pm$ 0.35)	1.08 ($\pm$ 0.35)	9.03 ($\pm$ 1.30)	2.81 ($\pm$ 1.20)
		0.6	2.41 ($\pm$ 0.07)	2.05 ($\pm$ 0.06)	1.20 ($\pm$ 0.39)	1.18 ($\pm$ 0.37)	9.19 ($\pm$ 1.45)	5.30 ($\pm$ 0.92)
		0.9	2.41 ($\pm$ 0.07)	1.38 ($\pm$ 0.04)	1.30 ($\pm$ 0.43)	1.28 ($\pm$ 0.40)	9.38 ($\pm$ 1.68)	7.72 ($\pm$ 0.48)